Week 1: Foundations of Ethical Hacking Task Documentation

This documentation is designed for the **YoungDevInterns_EthicalHacking_Tasks** repository. These notes provide a step-by-step guide for beginners to set up their environment, understand networking, and perform their first reconnaissance tasks.

1. Environment Setup

Before hacking, you need a controlled and legal environment. We use **Kali Linux** as our "Attacker" machine and **Metasploitable2** as our "Victim" (vulnerable) machine.

A. Installation

1. **Hypervisor:** Download and install **VMware Workstation Player** or **Oracle VirtualBox**.
2. **Kali Linux:** Download the "Pre-built Virtual Machine" version from kali.org. This is easier as it requires no manual installation.
3. **Metasploitable2:** Download the ZIP file from SourceForge. This is an intentionally vulnerable Linux machine.

B. Essential Linux Commands

In the terminal, you must be comfortable with these basics:

- `pwd:` Shows which folder you are currently in.
- `ls:` Lists all files and folders in the current directory.
- `cd [folder_name]:` Moves you into a specific folder.
- `sudo su:` Switches you to the "root" (Administrator) user.
- `mkdir [name]:` Creates a new folder.
- `ifconfig:` Displays your IP address and network interface details.

C. Creating the Secure Lab

To ensure your "hacking" doesn't leak into the real world:

1. Open your VM settings (VirtualBox/VMware).
2. Set the Network Adapter for **both** Kali and Metasploitable2 to **"Host-Only Adapter"** or a **"NAT Network."**
3. This creates a private "sandbox" where the two machines can talk to each other, but you won't accidentally scan devices on your actual home Wi-Fi.

2. Networking Basics

Ethical hacking is 90% networking. You cannot break into a system if you don't know how it communicates.

Core Concepts

- **IP Address:** The digital "home address" of a device (e.g., `192.168.1.5`).
- **MAC Address:** The physical hardware ID of a network card.
- **DNS (Domain Name System):** Translates `google.com` into an IP address.
- **TCP/IP:** The set of rules that governs how data is sent over the internet.
- **Ports:** Think of these as "doors" to a computer. Web servers usually use port 80 (HTTP) or 443 (HTTPS).

Network Discovery Tools

Open your Kali terminal and try these:

1. `ping [IP Address]:` Checks if a machine is "alive" and responding.
2. `traceroute [URL]:` Shows the path data takes to reach a destination.
3. `netstat -ano:` Shows all active connections on your own machine.

3. Footprinting and Reconnaissance

Reconnaissance (Recon) is the process of gathering information about a target.

Passive vs. Active Recon

- **Passive Recon:** Gathering info without interacting with the target (e.g., searching Google or LinkedIn). The target doesn't know you are watching.
- **Active Recon:** Interacting with the target system (e.g., scanning their ports). This can be detected by firewalls.

Essential Recon Tools

- **Whois:** Reveals who owns a domain and their contact info.
  - *Example:* `whois google.com`
- **nslookup:** Finds the IP address associated with a domain.
- **TheHarvester:** A tool to find emails, subdomains, and names related to a company.
- **Shodan.io:** A search engine for internet-connected devices (webcams, servers, etc.).

4. Hands-On Task: Nmap & Wireshark

This is where you apply what you have learned by scanning your local lab.

Step 1: Discovering the Target

Once your Metasploitable2 VM is running, login (username: `msfadmin`, password: `msfadmin`) and type `ifconfig` to find its IP address. Let's assume it is `192.168.56.101`.

Step 2: The Nmap Scan

Go to your Kali machine and run a basic scan to see which "doors" (ports) are open:

```
nmap -sV 192.168.56.101
```

- `-sV` tells Nmap to detect the version of the services running on the ports.
- **What to look for:** You will see a list of ports like 21 (FTP), 22 (SSH), and 80 (HTTP).

Step 3: Packet Analysis with Wireshark

1. Open **Wireshark** in Kali.
2. Select your network interface (usually `eth0` or `any`) and click the blue shark fin to start capturing.
3. While Wireshark is running, perform the Nmap scan again.
4. Stop the capture. Look for the **TCP** or **ARP** packets.
5. **The Goal:** Observe the "Three-Way Handshake" (SYN, SYN-ACK, ACK) that happens when Nmap talks to the victim machine.